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MHD numerical modelling of the plasma focus phenomena ☆

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Abstract

The motion of a current sheet in a plasma focus device has a complicated two-dimensional character. After the axial phase it rapidly changes to implosion and radial compression. Understanding of plasma evolution is impossible without computer simulation. In this paper results are presented of the computer simulation of current sheet dynamics in a plasma focus-1000 device. The MHD code is based on the modified free points method. This gridless method is very useful for the problem of large deformation. The physical model uses a full dissipative set of MHD equations with Braginski transport. The computations start from a partially ionized plasma and the kinetics of ionization is taken into account. In the region of low density, the anomalous resistivity due to the plasma turbulence is included. Current evolution is found from the circuit equation. Then numerical results differ a little from experimental results.

Keywords: Plasma focus device; MHD simulation; Z-pinch

1. Introduction

The plasma focus system (PF) was built in early 1960s as a modification of the classical Z-pinch by Filipov in Russia and Mather in the USA. During the discharge in the system of coaxial electrodes the current sheet, formed near the insulator, is moving toward the end of the electrode. Then the character of the movement is changed into a radial one. The hot and dense plasma (pinch) is formed on the axis after the compression phase. It is a source of powerful beams of electrons and ions as well as an intense source of X-rays and fusion neutrons (when the filling gas is deuterium). The beams and X-ray emission characteristics depend strongly on the plasma sheath evolution and dynamics. Their characteristics are relevant not only for basic plasma studies but also for different technology oriented experiments [1].

Diagnostics of these short-lived plasmas is a very difficult problem and despite many years of investigation there are many open questions [2].

The understanding of the physical process during the discharge in this device needs numerical simulations. They can be useful for planning further experiments.

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The dynamics of the PF has been calculated by many authors. The results of two-dimensional Magneto-Hydrodynamics (MHD) simulations of the PF have previously been reported by Potter [3], Dyachenko and Imshennik [4] and Maxon [5]. To obtain a good quantitative agreement with experimental results computer codes, which are based on a more sophisticated physical model are needed.

Research on the large PF-1000 facility in the Institute of Plasma Physics and Laser Microfusion has been done for a few years. The dynamics of the current sheet in our facility differs from the dynamics in other devices.

In this paper the results of the MHD simulation of the plasma dynamics in PF-1000 device have been presented. Our model includes, the full set of non-ideal MHD equations, and the kinetics of ionization. The influence of neutrals on transport coefficients and anomalous resistivity due to the growth of current instability have been taken into account.

2. Numerical method

The code is based on the modified free points method. This method was specially developed by Dyachenko for plasma focus simulation. There is no traditional computational mesh with the nodes rigidly connected with their neighbours. This stiff connection is the reason for mesh distortion in the Lagrange code. In the free points method each point has a set of adjacent points which are chosen in the physical neighbourhood, uniformly around the given one.

By the least-squares approximation of physical data for the neighbouring points the first derivatives have been found. The boundary points need special treatment to fulfill the boundary conditions. This has been done by using the Lagrange multipliers method to find the minimum in the least-squares method with a constriction on derivatives arising from boundary conditions. The second derivative has been found in a similar manner by approximation of the remainder after the first derivative computations.

All equations have been taken in a Lagrange frame and time advancing has been done explicitly.

The free points method is very useful for simulation of large deformation problems. One has to pay for the simplicity of the algorithm by a restriction of the time step due to the explicitly.

The non-conservative numerical scheme is another disadvantage. For the stabilizing algorithm a special procedure increasing the numerical diffusion is added.

3. Physical model

The main phase of the current sheet in PF evolution can be described in the frame of a non-ideal MHD model. It should be supplemented by taking into account atomic processes (i.e. ionization) in the neutral gas.

The magneto-hydrodynamics equations in this code are described in detail elsewhere [6]. They are solved for six unknown variables namely the mass density ρ, the radial and axial velocities u_r and u_z, the ion and electron temperatures T_i and T_e and the azimuthal magnetic field B_φ.

Transport coefficients are given according to Braginskii [6]. Therefore, we obtain the following system of equations:

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = -\rho\nabla\cdot\vec{u},$$

$$\rho\frac{\mathrm{d}\vec{u}}{\mathrm{d}t} = -\nabla p + \nabla\overleftrightarrow{\Pi} + \vec{j}\times\vec{B},$$

$$\rho c_{ve}\frac{\mathrm{d}T_e}{\mathrm{d}t} = -\nabla p_e \nabla\cdot\vec{u}_e - \nabla\cdot\vec{q}_e + \frac{1}{en_e}\vec{R}\cdot\vec{j} - Q_{e-i} - Q_{ioniz},$$

$$\rho c_{vi}\frac{\mathrm{d}T_i}{\mathrm{d}t} = -p_i\nabla\cdot\vec{u} + \overleftrightarrow{\Pi}_i\cdot\nabla\vec{u} - \nabla\cdot\vec{q}_i + Q_{e-i},$$

$$\frac{\mathrm{d}\vec{B}}{\mathrm{d}t} = \nabla\times\left[\vec{u}\times\vec{B} + \frac{1}{en_e}(\nabla p_e - \vec{R}) + \frac{1}{en_e}\vec{j}\times\vec{B}\right],$$

$$\vec{j} = \mu_0\nabla\times\vec{B},$$

where

$$\frac{d}{dt} \equiv \frac{\partial}{\partial t} + \vec{u} \cdot \nabla,$$

$$\vec{u} = (u_r,\ 0,\ u_z), \quad \vec{B} = (0,\ B_\varphi,\ 0), \quad \vec{j} = (j_r, 0, j_z),$$

$$p = p_e + p_i, \quad \overleftrightarrow{\Pi} = \overleftrightarrow{\Pi}_i + \overleftrightarrow{\Pi}_{art}, \quad v_e = v_{e-i} + v_{e-n},$$
$$v_i = v_{i-i} + v_{i-n}.$$

c_{ve}, c_{vi} are the electron and ion specific heats, respectively.

The system of MHD equations is closed by the circuit equation:

$$\frac{d}{dt}[L(t)I] + RI = U,$$

$$\frac{dU}{dt} = -\frac{I}{C}.$$

Apart from the ion velocity $\vec{u}$, we also need the electron (current) velocity

$$\vec{u}_e = \vec{u} - \frac{m_i\vec{j}}{\rho e}.$$

When the current velocity is high enough the current micro-instabilities can grow to the level which gives a high anomalous resistivity. The anomalous resistivity caused by lower hybrid and modified Buneman instabilities has been included in the model. In the equation for electron temperature, Joule heating, energy exchange with ions and ionization and radiation losses were included. The equation for ion temperature contains the energy exchange between ions and electrons, and viscous heating.

In the inelastic process it is necessary to include electron impact ionization from the ground state radiative recombination and three body recombination. Since ionization and recombination are functions of electron temperature and density, respectively, it is needed to solve the resulting equation

$$\frac{dn_e}{dt} = n_e(n_0 - n_e) \cdot S - \alpha_r n_e^2 - \beta_{3B} n_e^3$$

together with the energy equation for electron component.

Boundary conditions of a typical nature were taken:

(a) on the pinch surface

$$\vec{n} \cdot \vec{q}_i = 0, \quad \vec{n} \cdot \vec{q}_e = 0,$$

$$B|_{r=r_p} = \frac{I}{r_p},$$

$$\vec{n} \cdot \left(p\overleftrightarrow{I} - \Pi\right) = 0,$$

(b) on the electrode

$$u_n = 0,$$

$$\frac{\partial T_e}{\partial n} = 0, \quad \frac{\partial T_i}{\partial n} = 0,$$

(c) on the axis

$$u_r = 0, \quad B_\varphi = 0,$$

$$\frac{\partial T_e}{\partial r} = 0, \quad \frac{\partial T_i}{\partial r} = 0.$$

In our experimental device PF-1000 the outer electrode has been made as a rod arrangement. In the model a solid outer electrode has been assumed.

The calculations have started after the breakdown phase when the current sheet is formed. It was assumed that the current sheet starts to move from the end of insulator at position -0.5 m from the end of the inner electrode.

The parameters of PF-1000 facility have been taken for the simulations. They are as follows: radius of the inner electrode 0.12 m, outer electrode 0.18 m, hollow radius in the centre of the electrode 0.015 m, electrode length 0.60 m. The dimensions of the computational area on the end of the electrode has been restricted do the rectangular $r_m = 0.3$ m and $z_m = 0.26$ m.

Computations started from the room temperature $T_e = T_i = 300$ K, and low initial degree of ionization equal to 1%. The filling pressure of the deuterium gas was 400 Pa.

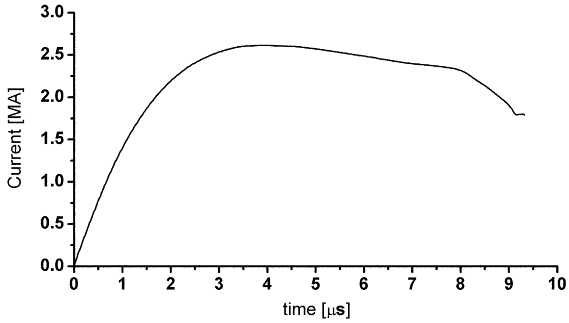

Fig. 1. The current versus time.

4. Results and discussion

The preliminary results of simulations agree qualitatively with experimental results [7]. The maximum current 2.3 MA (Fig. 1) obtained in the computation is about 15% greater than the measured one. The velocity computed for the run down phase is greater than the measured one: $u_{z\,\mathrm{exp}} = 8 \times 10^4\,\mathrm{m/s}$; $u_{z\,\mathrm{comp}} = 8.9 \times 10^4\,\mathrm{m/s}$.

Number density profiles at various times are shown in Fig. 2. Note that the plasma sheet is wide and the density has a parabolic profile. Near the axis for the region of minimal radius of plasma column equal to 0.048 m, the density is equal to $5 \times 10^{25}\,\mathrm{m}^{-3}$.

The initial sheet velocity in the collapse phase is rather low. Only when the sheet is near the axis it rapidly grows up to $4 \times 10^4\,\mathrm{m/s}$. There is insufficient experimental data in this phase to compare with computations.

The shock wave reaches the pinch axis after $8.5 \times 10^{-6}\,\mathrm{s}$ and the ion temperature grows up to 1.2 keV in this moment. We can see from Fig. 3 that maximum value of electron temperature is lower and equal to about 600 eV. There is no evidence of $R-T$ instability in the compression phase. The current-sheet curvature and the axial plasma flow stabilized the plasma boundary. One can see from Fig. 2c, that a minimum radius has been reached 0.15 mm from the anode face.

Simulations were stopped at the moment of current sheet stagnation near the axis when the MHD model is not valid.

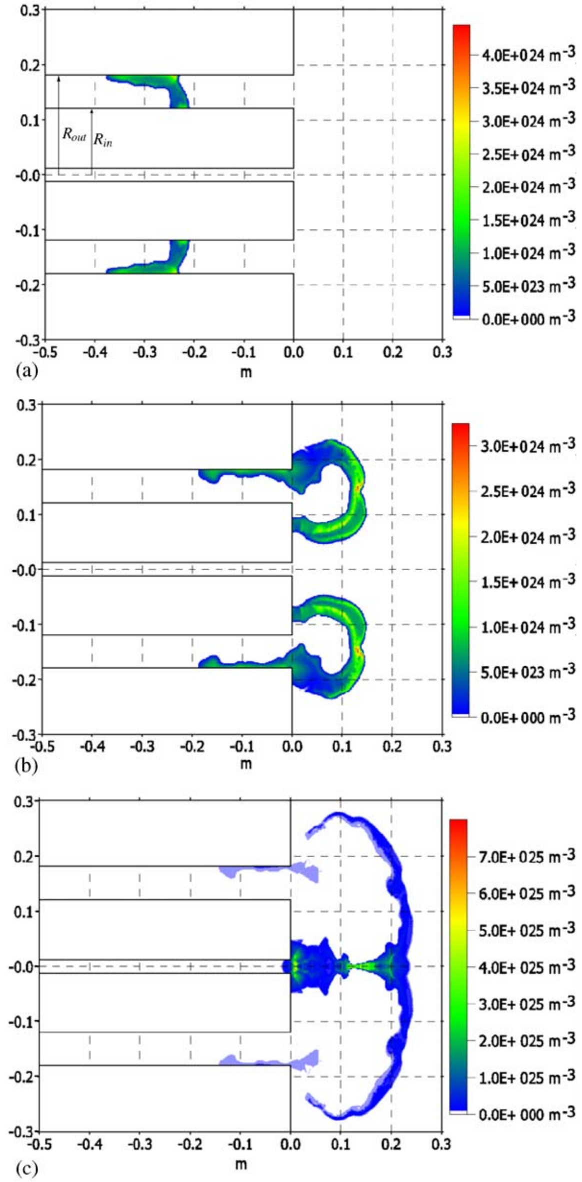

Fig. 2. Contours of number density profiles: (a) for $t = 4\,\mu\mathrm{s}$; (b) for $t = 9\,\mu\mathrm{s}$; (c) for $t = 10\,\mu\mathrm{s}$.

5. Conclusions

The results of studies described in this paper can be summarized as follows.

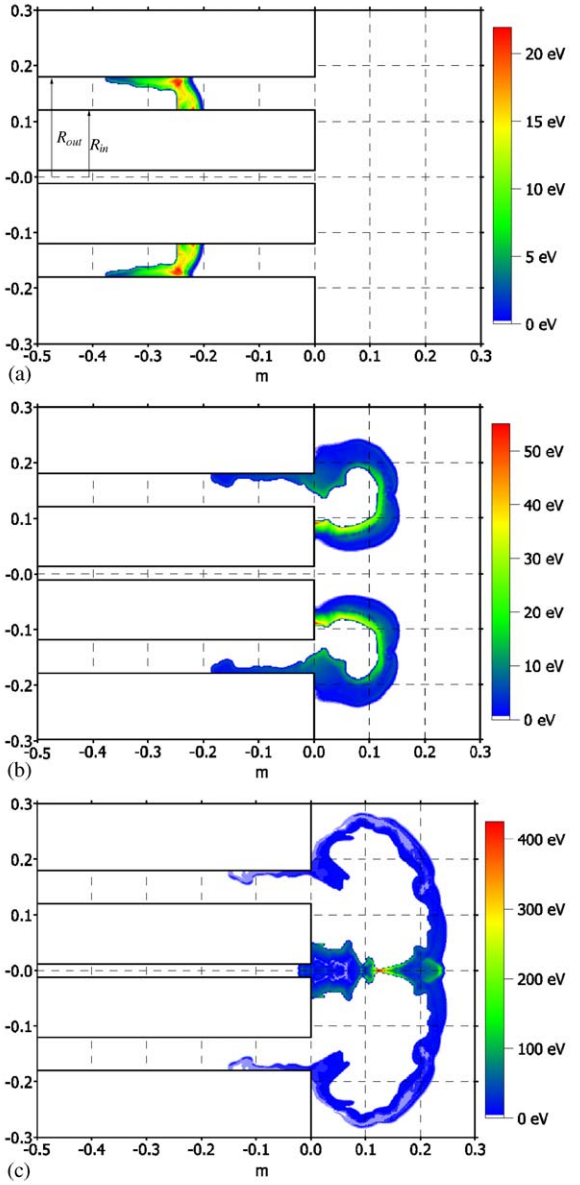

Fig. 3. Contours of electron temperature: (a) for $t = 4\,\mu s$; (b) for $t = 9\,\mu s$; (c) for $t = 10\,\mu s$.

(1) A stable plasma structure situated between two pinching regions is seen.
(2) The specific features of the PF-1000 phenomena; radial dynamics with two separate structures positioned along the axis are shown. The reason for this is the longer pinch length resulting from the larger bank energy and bigger electrode dimensions.
(3) Qualitative agreement of the experimentally-observed pinch dynamics and structures [8] with numerical modelling may be observed.

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